

0 8 . 1

The *KRAS* gene codes for a protein called K-Ras. The protein relays signals from outside a cell to a cell's nucleus, stimulating cell division. An alteration in the *KRAS* gene produces an oncogene which can cause a tumour to develop.

Suggest and explain how an alteration in the *KRAS* gene can cause a tumour to develop.

[3 marks]

0 8 . 2

Alterations in the *KRAS* gene can cause colorectal cancer (CRC). Scientists investigated the survival time of the following three groups of CRC patients who had type II diabetes.

Group **A** – received no drug to treat type II diabetes

Group **B** – received the drug metformin to treat type II diabetes

Group **C** – received a combination of drugs (including metformin) to treat type II diabetes

The scientists used a statistical test to compare the survival time of these patients with CRC patients with no history of type II diabetes.

Turn over for the next question

Turn over ►

Group	Mean survival time of CRC patients / months	Probability value (P)
No history of type II diabetes	32.2	–
A	21.3	0.007
B	49.7	0.022
C	38.3	0.636

